



coal-seam fires

what are they and why i care about them

what are they?

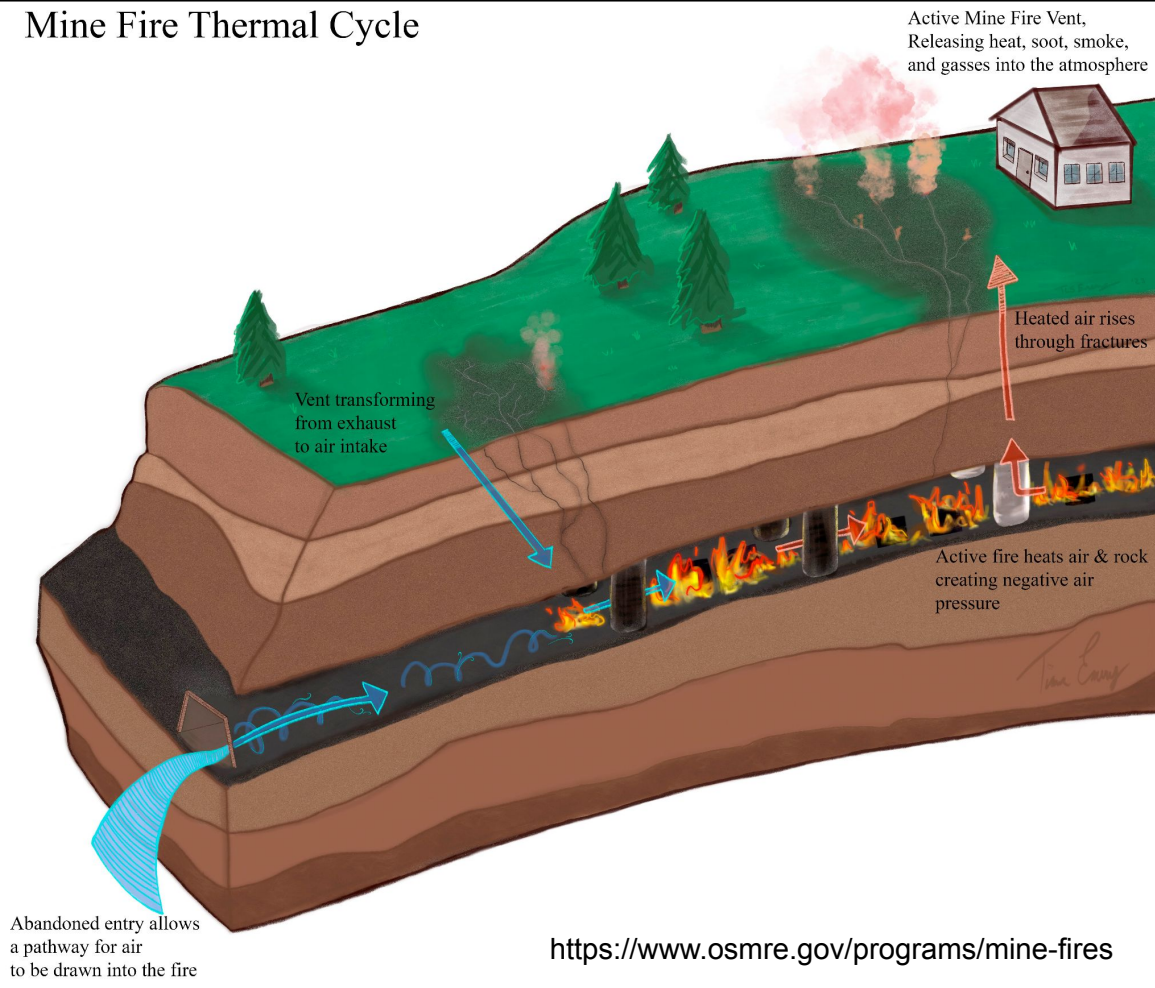
- persistent fires in underground natural coal seams or man-made coal mines
- can last 1000's of years
- there are many burning right now!

what causes them?

- the majority of coal-seam fires:
 - start within active or abandoned coal mines
 - start through careless human activity
 - burning debris
 - accidental gas explosions
- can also start naturally
 - forest fires
 - lightning strikes
 - lava flows

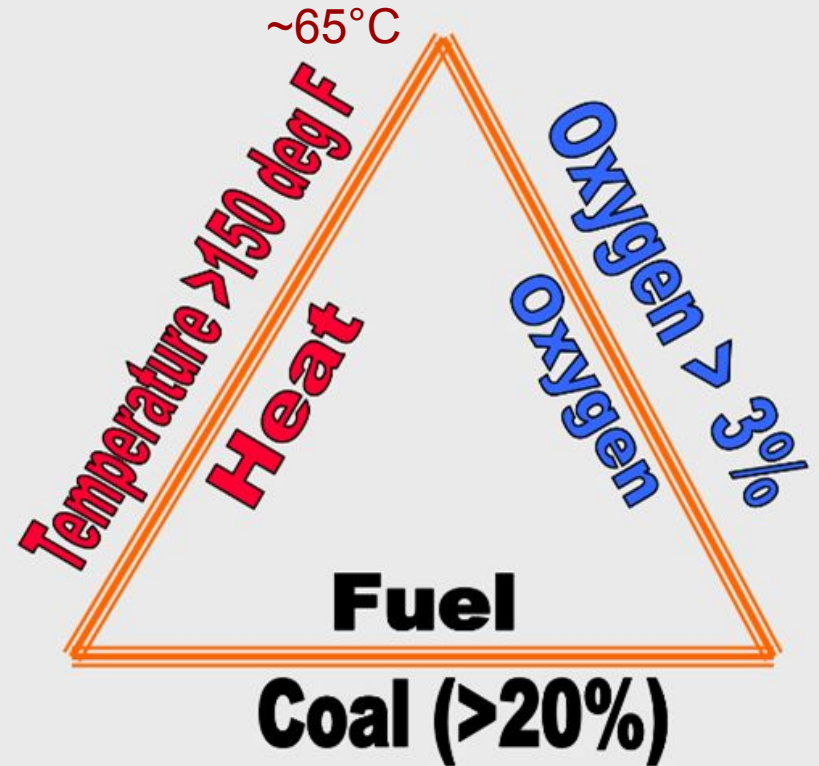
life cycle?

Mine Fire Thermal Cycle



how are they extinguished?

remove 1 leg from the “fire triangle”



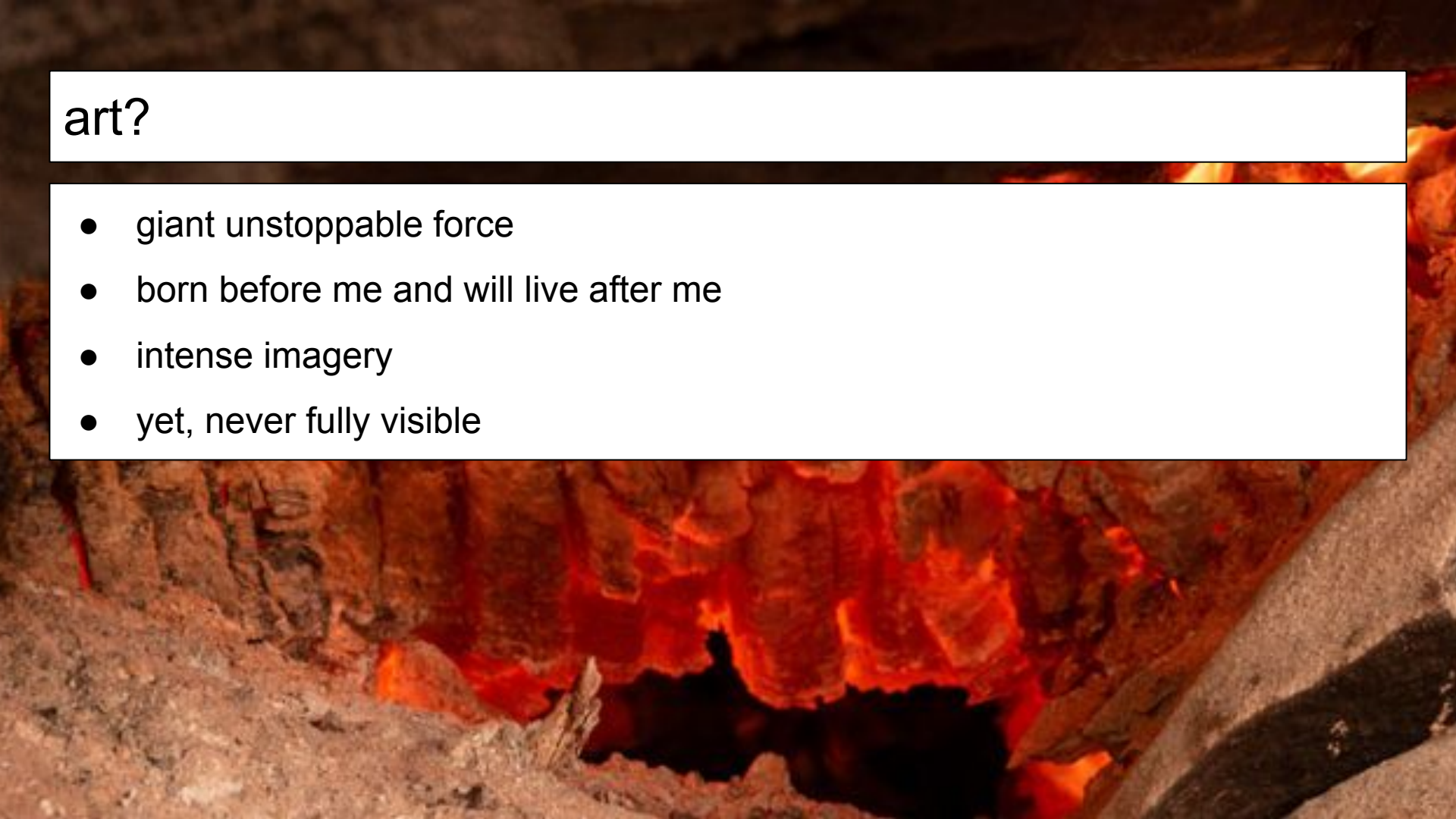
examples?

- Wingen, New South Wales, Australia
 - has burned for approximately 6,000 years (estimated)
 - oldest known coal fire
- Centralia, Pennsylvania, United States
 - may 27, 1962
 - could continue to burn for over 250 years
- Australia, Canada, China, France, Germany, India, Indonesia, New Zealand, Norway, South Africa, United States
- <https://youtu.be/XjT81UGoxdU?si=Cen4HfXwdAKyPMsn&t=75>

https://en.wikipedia.org/wiki/Coal-seam_fire
https://en.wikipedia.org/wiki/Burning_Mountain
https://en.wikipedia.org/wiki/Centralia_mine_fire

why do i care about them?

- environmental impact
- China's coal fires
 - Consume ~20 – 200 million tons of coal a year
 - 1 percent of the global carbon dioxide emissions from fossil fuels
- byproducts and off-gasses are bad for local communities (plants, animals, humans, etc)



art?

- giant unstoppable force
- born before me and will live after me
- intense imagery
- yet, never fully visible